Question 1: Decision Tree (AND Function) [CLO-1]

Consider a binary classification problem with inputs:

$$(x_1, x_2, x_3) \in \{0,1\}^3$$

and output:

$$y = x_1 \land x_2 \land x_3$$

Construct a decision tree using **Information Gain**. Clearly justify the choice of the root node and present the final tree.

Question 2: Gradient Descent on Quadratic Loss [CLO-1]

Consider the loss function:

$$L(w) = (w - 3)^2 + 2$$

with initial value $w_0 = 0$and learning rate $\eta = 0.1$.

Perform two iterations of gradient descent and report the updated values of $w$and the corresponding loss at each step. Comment on the behavior of the optimization process.

Question 3: K-Means Clustering [CLO-2]

You are given the dataset:

$$X = \{(1,1), (2,1), (4,3), (5,4)\}$$

with initial centroids:

$$\mu_1 = (1,1), \mu_2 = (5,4)$$

Perform two iterations of K-means clustering and report the final cluster assignments and centroids.